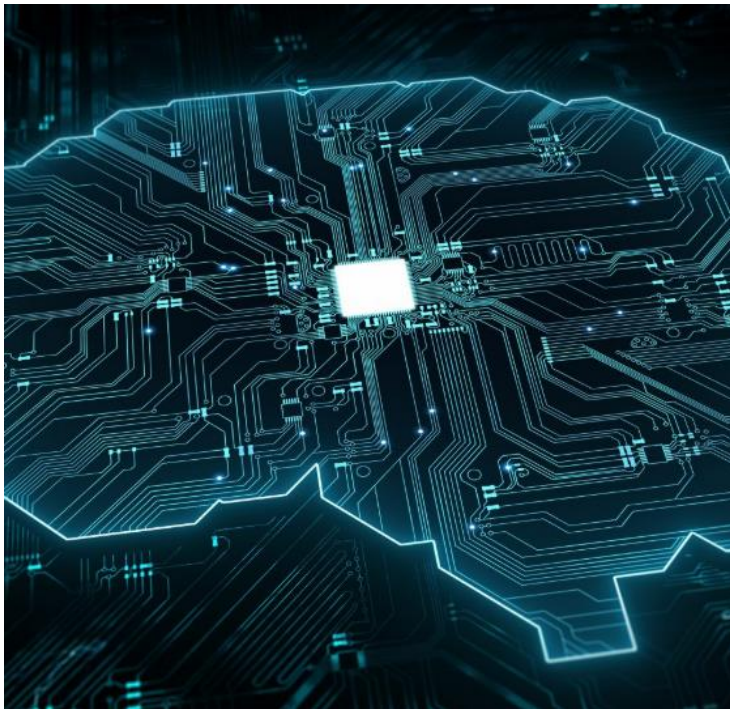




MCU 是什麼？

/ 先懂 MCU 是什麼



接收資訊 (眼睛、耳朵)

思考判斷(MCU)

發出指令 (手腳動作)

/ MCU 都用在哪裡？



🛒 洗衣機 (控制
水位、轉速)



🍲 電鍋、微波
爐



🚲 電動腳踏車



🚗 汽車 (車窗、
雨刷、引擎控制)

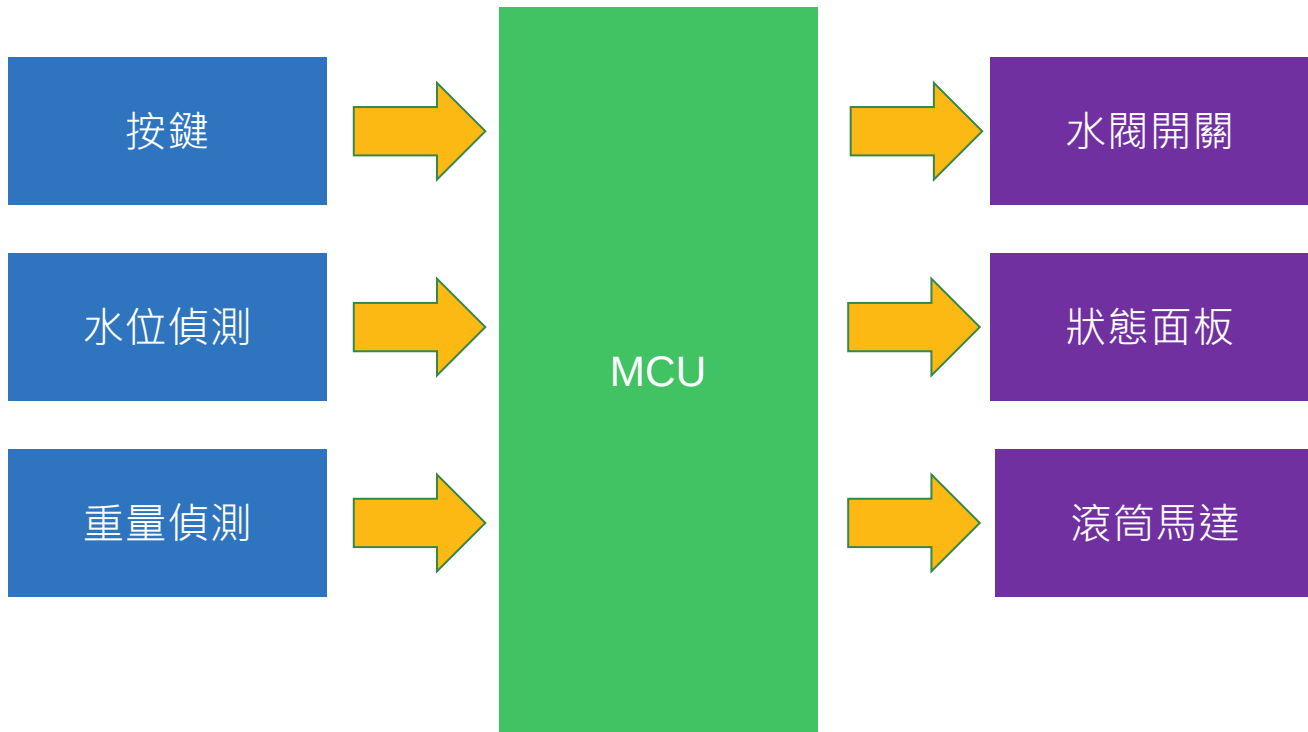


📱 智慧手錶、穿戴裝置



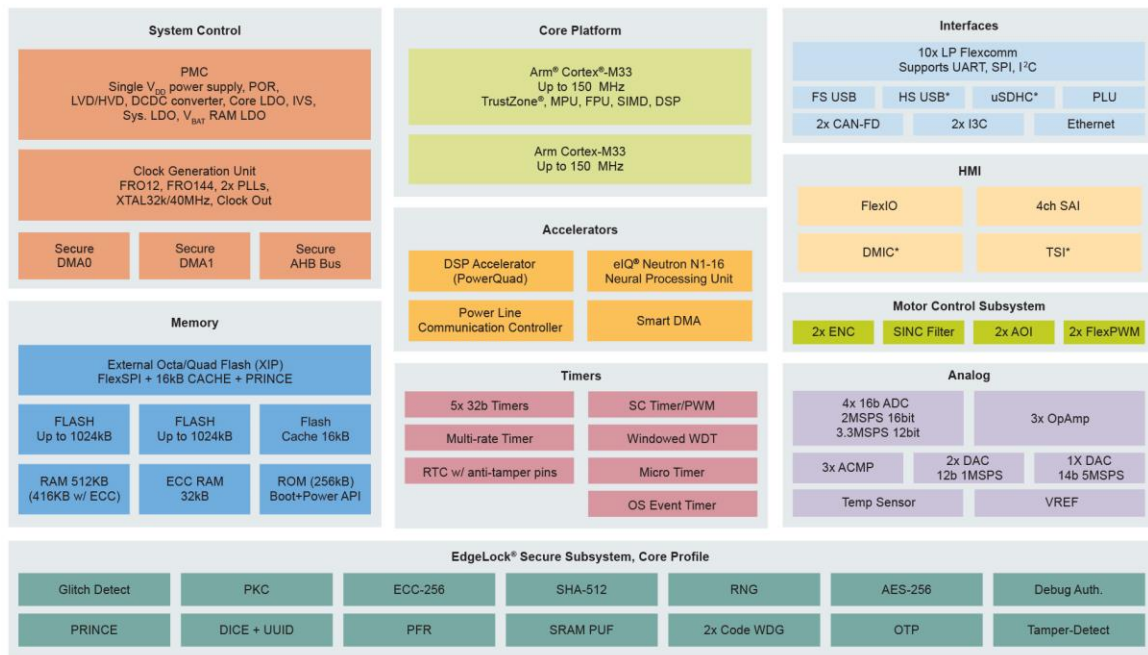
🤖 機器人

/ 洗衣機基本控制



Get To Know MCXN947

Key Features and Benefits



Key Feature Highlights

Performance

- Dual CM33 150MHz
- DSP, NPU

Intelligent Peripherals

- Flexcomm, FlexIO, FlexPWM, FlexSPI
- Touch, Motor Control, Analog

Advanced Security

- Crypto, Tamper-Detect
- Life-cycle management

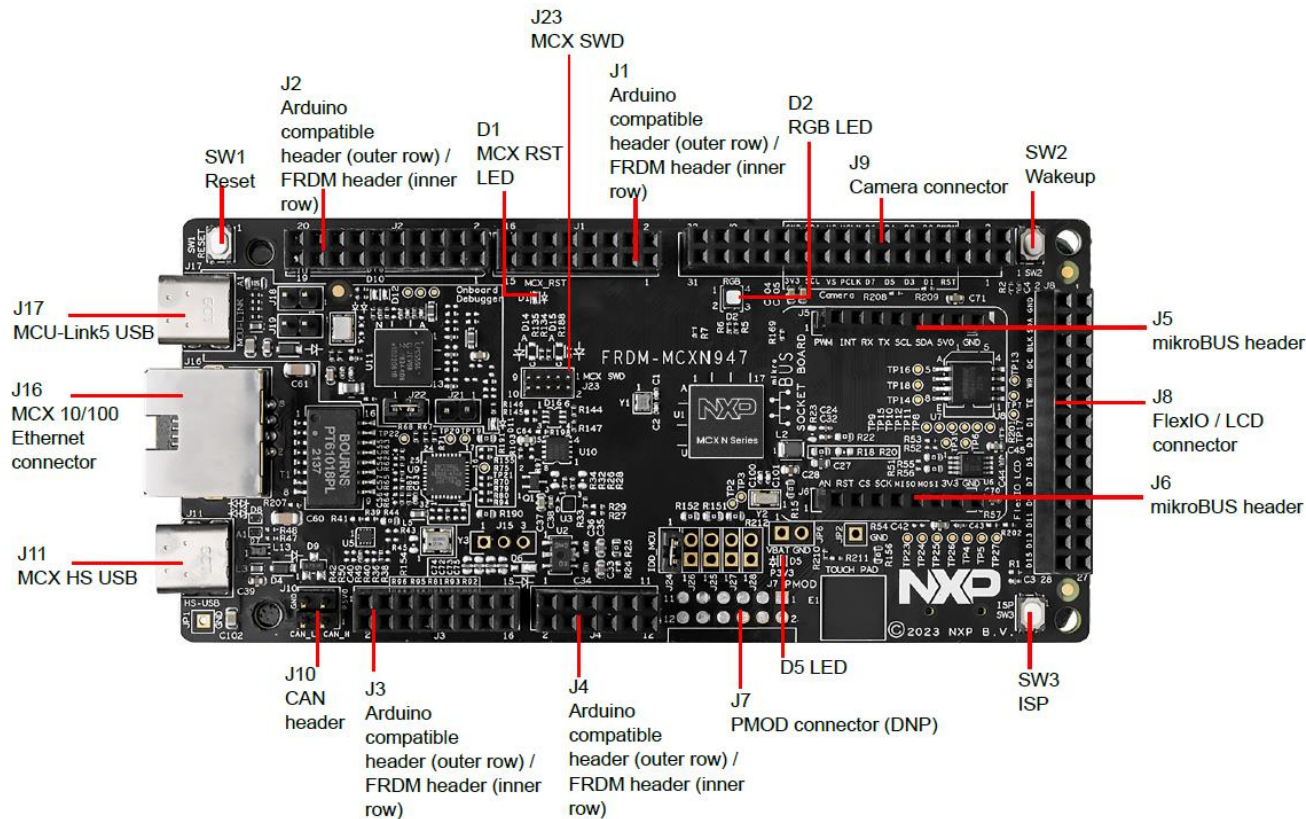
Power and Operation

- Buck DC-DC, Core LDO, other LDOs
- Separate AO domain on VBAT pin
- Operating voltage: 1.71V to 3.6V
- Temperature range: -40 to 105 °C
- IOs: 1.71V to 3.6V

Package Options

- VFBGA184: 9 x 9 x 0.86 mm, 0.5 mm pitch
- HLQFP100: 14 x 14 x 1.4 mm, 0.5 mm pitch

FRDM-MCXXN947



開發工具

MCX UNIFIED DEVELOPER EXPERIENCE

採用 MCUXpresso 軟體與工具套件

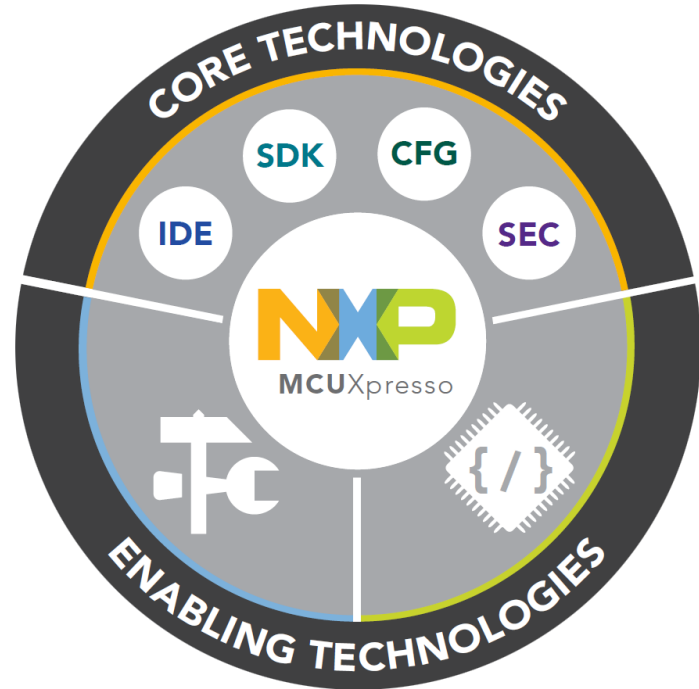
完整的 MCUXpresso SDK

多樣化的 RTOS

完整的驅動程式與中介軟體支援來自 NXP 及其合作夥伴生態系的更多選擇

UNIFIED SOFTWARE
AND TOOLS

SUPPORTED BY MCUXpresso



Flexible and Rapid Development for MCUXpresso

Hardware Platforms

FRDM Development Boards

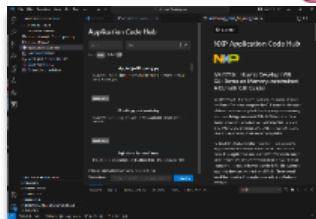
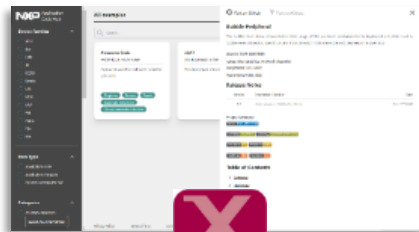


- Arduino® Header
- FRDM Header
- mikroBUS™ Header
- Pmod™ Header

- Standard form factor and headers
- Easy access to MCU I/Os
- On-board MCU-Link debugger

Easy Software Integration

- Application middleware from NXP and Open-CMSIS-Packs
- Easily find software examples and demos from your browser or directly in NXP IDEs using [Application Code Hub](#)



Rapid Application Development

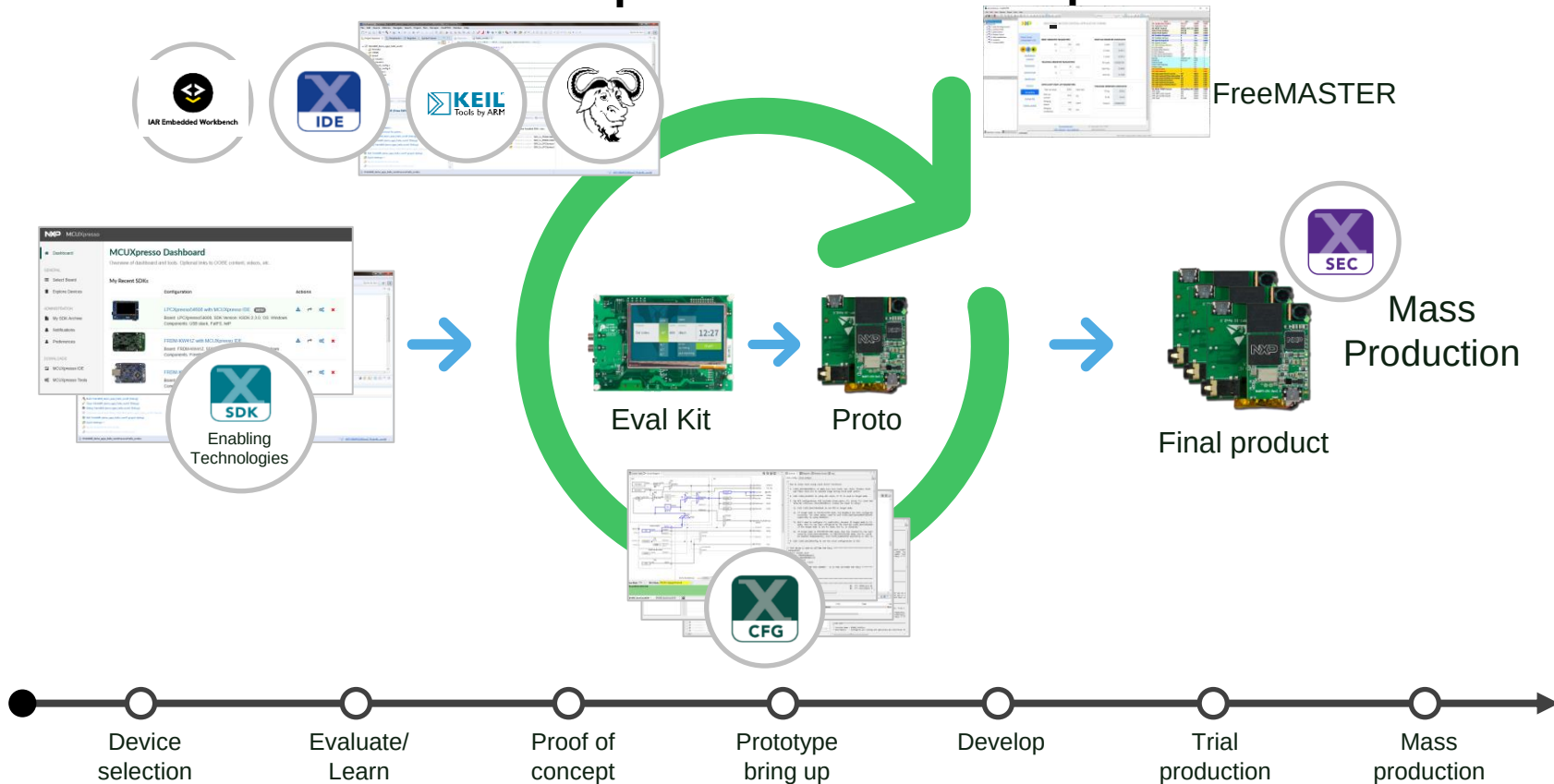
- Broad range of supported modules
- Shield boards come with standard, pre-populated headers
- [Expansion Board Hub](#) to find add-on boards from NXP and our partners with related MCUXpresso SDK-compatible drivers and examples
- Development for Connectivity, Graphics, Machine Learning, Motor Control, Sensors, Touch and Voice



Full-Featured Evaluation Kits



Streamlined MCUXpresso development flow



MCUXpresso SDK

Software framework and drivers

Architecture:

- CMSIS-CORE compatible
- Single driver for each peripheral
- Transactional APIs w/ optional DMA support for communication peripherals

Integrated RTOS options:

- Azure RTOS ThreadX
- FreeRTOS
- RTOS-native driver wrappers

Enabling SW Technologies:

- Wired Connectivity (Ethernet, TSN)
- Wireless Connectivity (WiFi, Bluetooth, Thread, Matter, Zigbee)
- HMI (Graphics, Voice, Touch)
- Audio
- Motor Control
- eIQ Machine Learning
- Cloud connectivity
- Accelerators and specialized peripherals
- Safety
- Security
- Storage

Reference Software:

- Peripheral driver usage examples
- Application demos
- Azure RTOS and FreeRTOS usage demos
- IoT connectivity examples

License:

- BSD 3-clause for startup, drivers, USB stack
- All code Black Duck scanned

Download options:

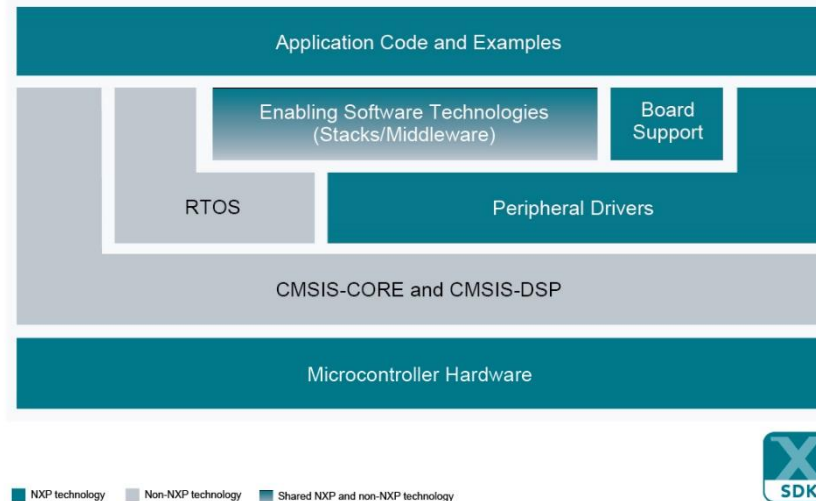
- Custom SDK web builder for optimized download size
- Direct download within MCUXpresso
- GitHub repository leveraging community engagement
- CMSIS Packs for supported toolchains

Toolchains:

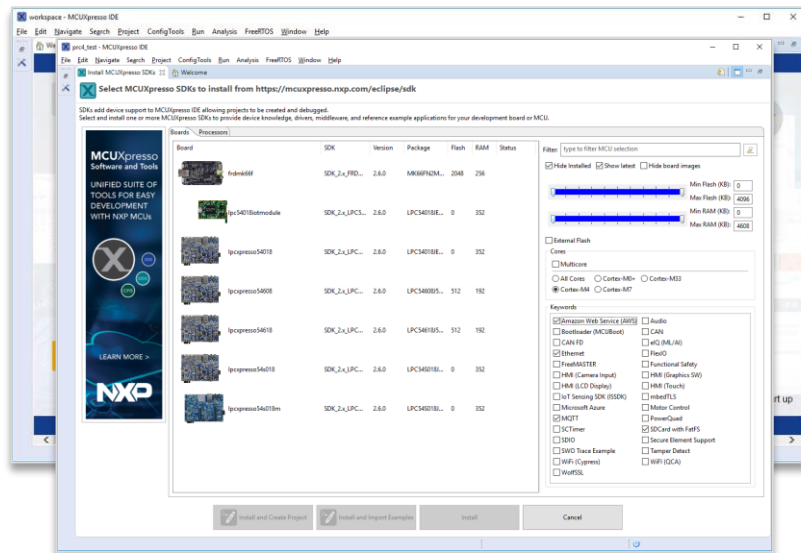
- MCUXpresso IDE
- IAR®, ARM® Keil®, GCC w/ Cmake

Quality:

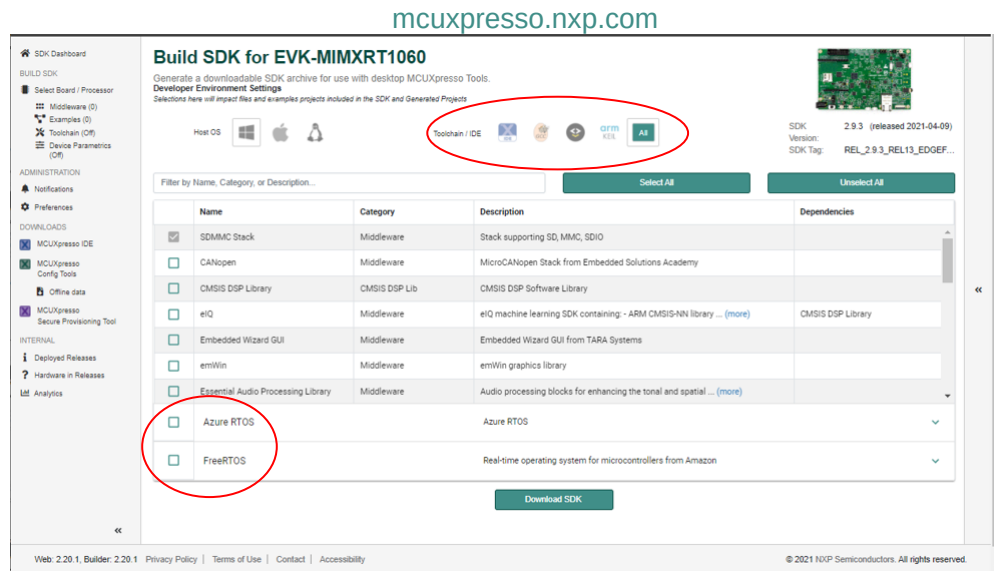
- Production-grade software
- Consistent API architecture across all releases
- MISRA 2012 compliance
- Checked with Coverity® static analysis tools
- Extensive Continuous Integration testing across range of compatible development boards, supported toolchains, and debug probes



SDK Selection for Board or Device

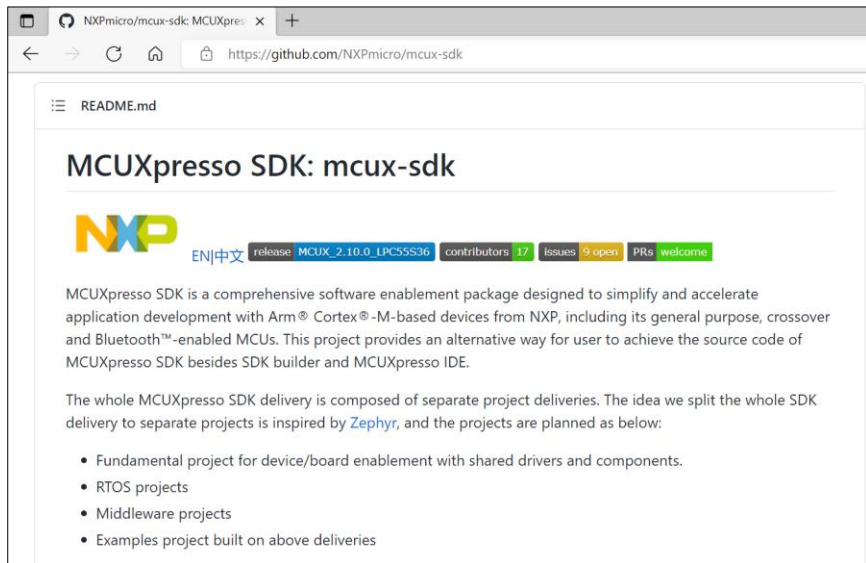


Simple selection of pre-built SDKs from within MCUXpresso IDE



Custom SDK package builder available online
Support MCUXpresso IDE, GCC, IAR & Keil
Options for Azure RTOS ThreadX or FreeRTOS

SDK available from github



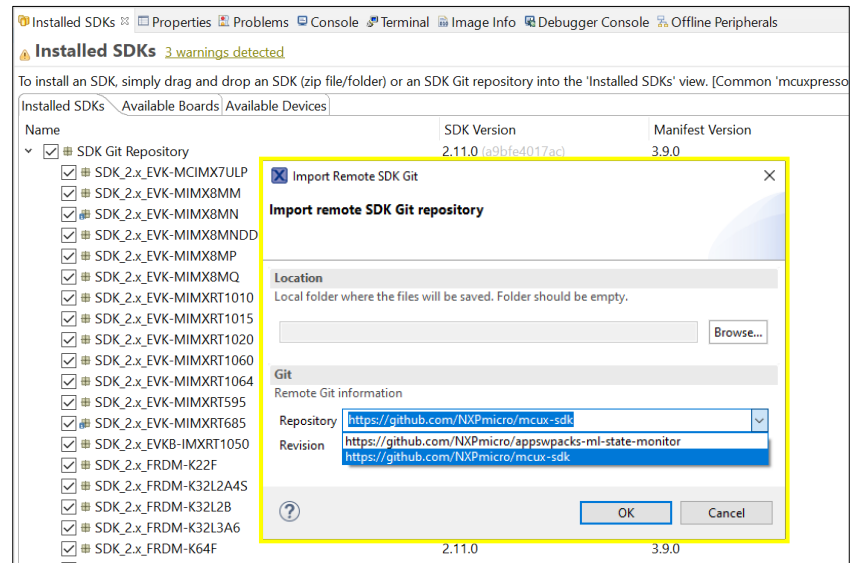
MCUXpresso SDK: mcux-sdk

EN|中文 release MCUX_2.10.0_LPC55536 contributors 17 issues 9 open PRs welcome

MCUXpresso SDK is a comprehensive software enablement package designed to simplify and accelerate application development with Arm® Cortex®-M-based devices from NXP, including its general purpose, crossover and Bluetooth™-enabled MCUs. This project provides an alternative way for user to achieve the source code of MCUXpresso SDK besides SDK builder and MCUXpresso IDE.

The whole MCUXpresso SDK delivery is composed of separate project deliveries. The idea we split the whole SDK delivery to separate projects is inspired by Zephyr, and the projects are planned as below:

- Fundamental project for device/board enablement with shared drivers and components.
- RTOS projects
- Middleware projects
- Examples project built on above deliveries



Installed SDKs 3 warnings detected

To install an SDK, simply drag and drop an SDK (zip file/folder) or an SDK Git repository into the 'Installed SDKs' view. [Common 'mcuxpresso' Installed SDKs / Available Boards / Available Devices]

Name	SDK Version	Manifest Version
SDK Git Repository	2.11.0 (a9bfe4017ac)	3.9.0
SDK_2.x_EVK-MCIMX7ULP		
SDK_2.x_EVK-MIMX8MM		
SDK_2.x_EVK-MIMX8MN		
SDK_2.x_EVK-MIMX8MNDD		
SDK_2.x_EVK-MIMX8MP		
SDK_2.x_EVK-MIMX8MQ		
SDK_2.x_EVK-MIMXRT1010		
SDK_2.x_EVK-MIMXRT1015		
SDK_2.x_EVK-MIMXRT1020		
SDK_2.x_EVK-MIMXRT1060		
SDK_2.x_EVK-MIMXRT1064		
SDK_2.x_EVK-MIMXRT595		
SDK_2.x_EVK-MIMXRT685		
SDK_2.x_EVKB-IMXRT1050		
SDK_2.x_FRDM-K22F		
SDK_2.x_FRDM-K32L2A4S		
SDK_2.x_FRDM-K32L2B		
SDK_2.x_FRDM-K32L3A6		
SDK_2.x_FRDM-K64F		

Import Remote SDK Git

Import remote SDK Git repository

Location
Local folder where the files will be saved. Folder should be empty.
Browse...

Git
Remote Git information

Repository: <https://github.com/NXPmicro/mcux-sdk>

Revision: <https://github.com/NXPmicro/appswpacs-ml-state-monitor>
<https://github.com/NXPmicro/mcux-sdk>

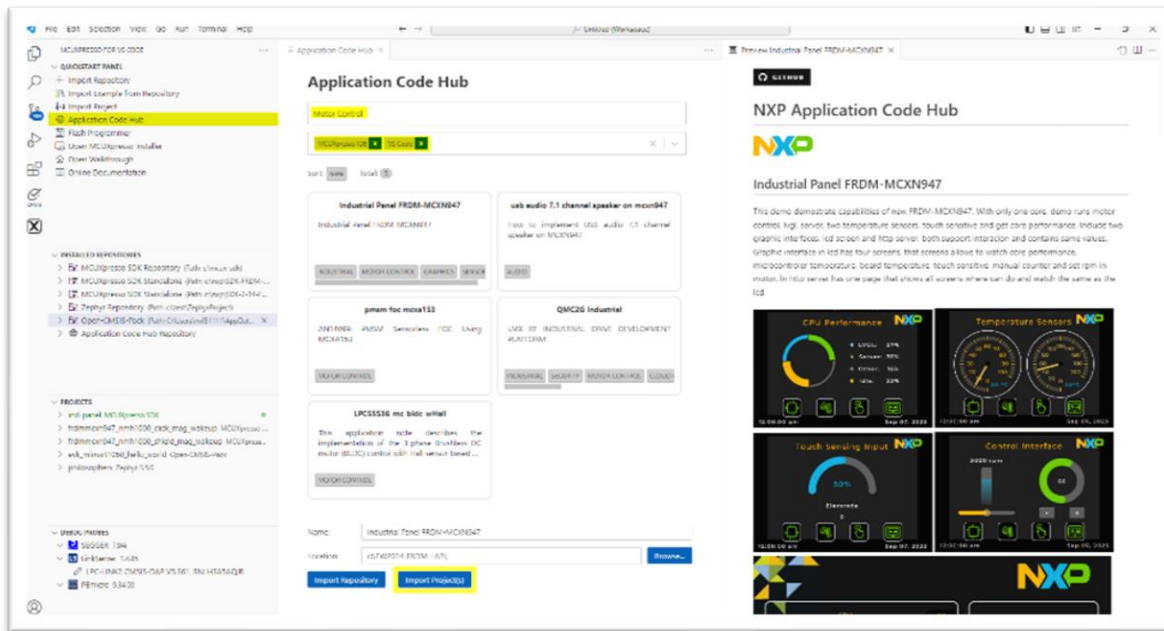
OK Cancel

<https://github.com/NXPmicro/mcux-sdk>

- Remote repository with multiple SDKs

NXP 資源

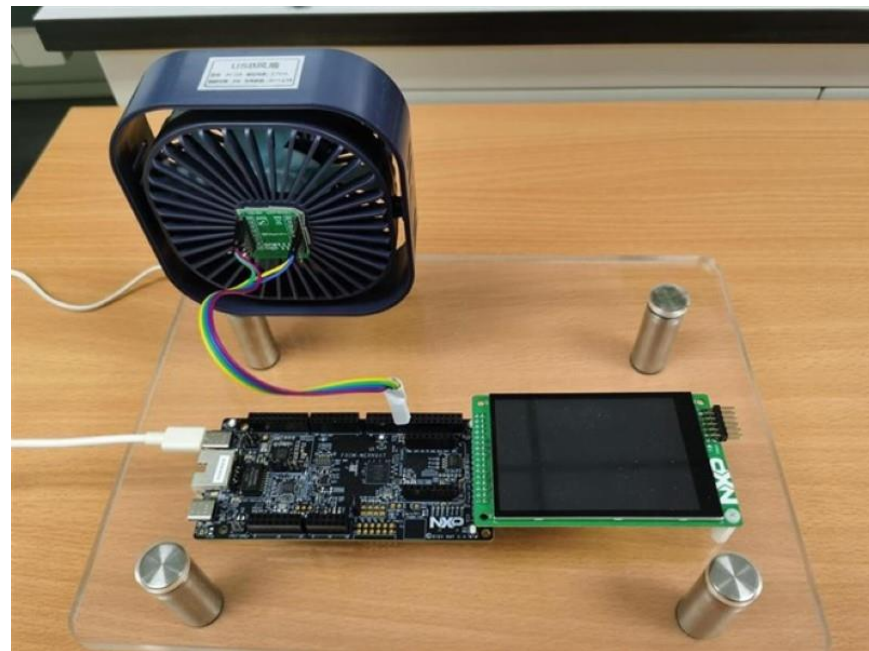
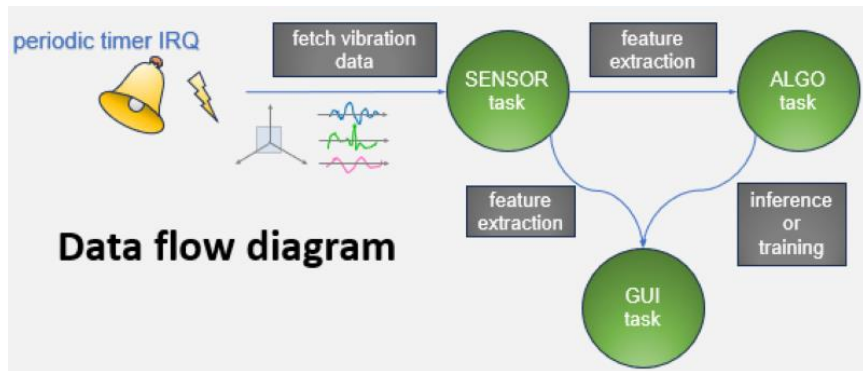
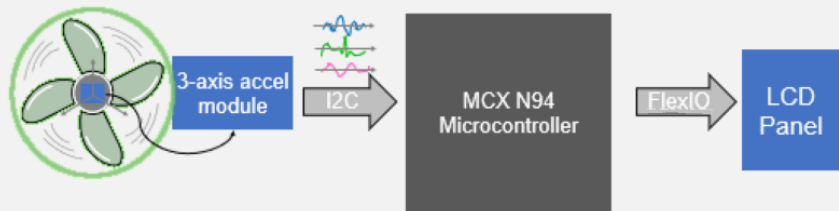
Application Code Hub



<https://mcuxpresso.nxp.com/appcodehub?deviceFamily=mcx&search=MCXN9>

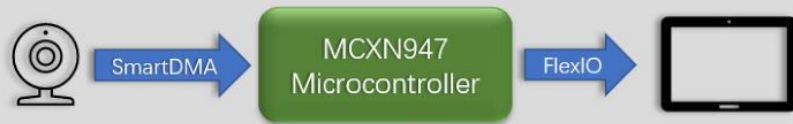
Anomaly detection on MCXN947

Block diagram

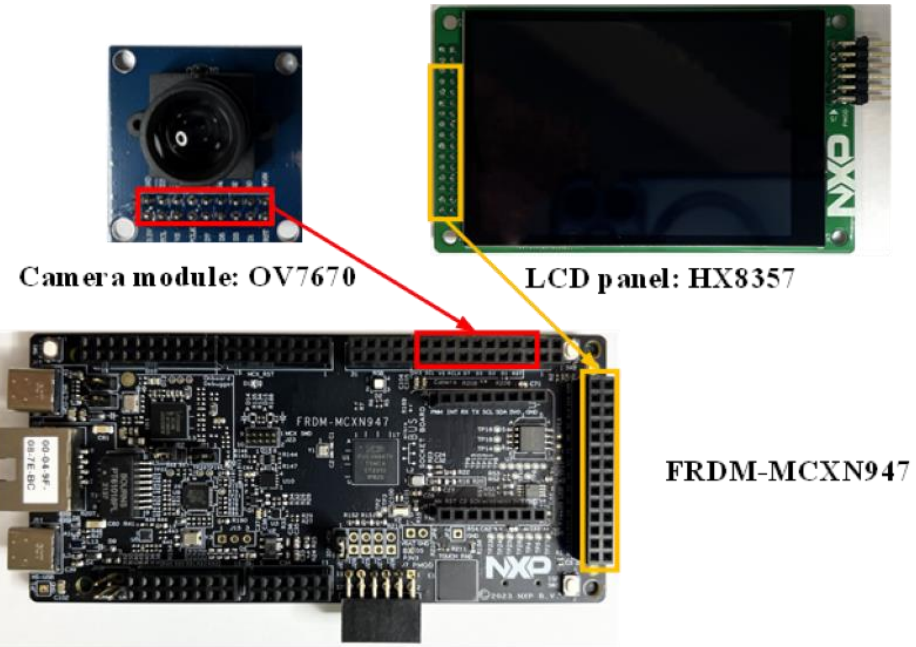
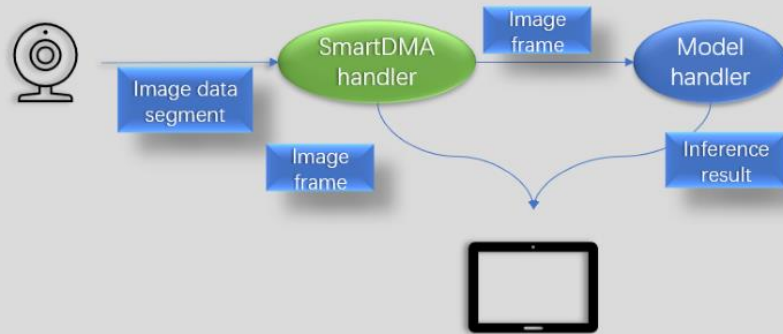


/ Multiple Face Detection on

Block diagram



Data flow diagram



/ Doom game to run on NXP MCX



License **GPL 2.0** Language **C** Board **FRDM-MCXN947** Category **RTOS** Category **HMI** Category **GRAPHICS**

/ Doom game to run on NXP MCX



Thank You